DATA ANALYTICS CAPSTONE PROJECT

Bluestock Mutual Fund Analytics & Investor Insights Platform

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Domain: Mutual Fund Analytics

Technology Stack

Python · Pandas · NumPy · SciPy · PostgreSQL

Power BI · Matplotlib · Jupyter Notebook · GitHub

Contents

Chapter 1

Executive Summary

The **Bluestock Mutual Fund Analytics & Investor Insights Platform** is a full-spectrum data analytics capstone project that delivers institutional-grade analysis of the Indian mutual fund industry. The project ingests, cleans, and transforms multi-dimensional datasets encompassing industry-level AUM and SIP statistics, scheme-level NAV histories, fund performance metrics, synthetic investor transaction records, and market benchmark data. It then applies rigorous quantitative finance methodologies and modern business intelligence techniques to generate actionable insights for investors, fund managers, and wealth advisors.

1.1 Analytical Approach

The project follows a structured pipeline: data engineering and ETL (Extract-Transform-Load) in Python/PostgreSQL, exploratory data analysis with Matplotlib, performance analytics computing Sharpe Ratio, Sortino Ratio, Alpha, Beta, CAGR, and Maximum Drawdown for every fund, advanced analytics encompassing Value-at-Risk, Conditional VaR, rolling Sharpe analysis, Herfindahl-Hirschman Index for sector concentration, SIP continuity modeling, and investor cohort segmentation, culminating in a four-page interactive Power BI dashboard.

1.2 Major Findings

The Indian mutual fund industry's total Assets Under Management reached **INR 62.74 lakh crore**, with SIP monthly inflows at **INR 31,000 crore** and **26.12 crore folios**. The composite fund scorecard ranked fund **120505** (AMFI code) at the top with a perfect score of **100.00**, an **18.08%** three-year return, a Sharpe Ratio of **1.18**, and Alpha of **0.30**. SBI Mutual Fund leads the industry with over INR 13 lakh crore in AUM, followed by ICICI Prudential and HDFC Mutual Fund.

Among 40 schemes analyzed, annualized standard deviations ranged from **13.74%**

to **25.79%**, revealing substantial risk dispersion. The Liquid category dominated net inflows with nearly **INR 4,51,275 crore** over 2024–2025, while SIP inflows versus the Nifty 50 exhibited strong co-movement, rising from approximately INR 0.13M crore in 2022 to over INR 0.35M crore by 2025.

1.3 Dashboard Development

The Power BI dashboard comprises four interactive pages — *Industry Overview*, *Fund Performance*, *Investor Analytics*, and *SIP & Market Trends* — each equipped with KPI cards, dynamic filters, drill-through navigation, and comparative visualizations.

1.4 Business Value

The platform provides a replicable, scalable analytics framework that transforms raw financial data into structured intelligence. Retail investors gain a transparent fund scoring mechanism, fund managers receive risk-decomposition insights, and wealth advisors obtain investor-behavior analytics to improve client retention and portfolio construction.

Chapter 2

Industry Background

The Indian mutual fund industry has undergone a structural transformation over the past decade, evolving from a niche investment vehicle into a mainstream wealth-creation channel for millions of retail and institutional investors. Several macro-level trends underpin this growth.

2.1 AUM Expansion

Total industry AUM has grown significantly over the 2022–2025 period. The AUM Growth Trend analysis demonstrates a clear upward trajectory, with AUM rising steadily from below INR 90 lakh crore in early 2022 to a peak of approximately **INR 150 lakh crore** by mid-2024, followed by a correction to approximately INR 120 lakh crore by early 2025. This pattern mirrors India's broader equity market cycle — a strong bull run through 2023–2024, driven by domestic liquidity and retail participation, followed by profit-booking and global macroeconomic headwinds.

2.2 SIP Revolution

Systematic Investment Plans have become the primary channel for retail participation. The analysis captures SIP monthly inflows at **INR 31,000 crore**, with **9.35 crore** active SIP accounts and a latest SIP AUM of **INR 15.90 lakh crore**. SIP inflows have grown from approximately INR 0.13M crore in 2022 to over INR 0.35M crore in 2025, even as the Nifty 50 moved from approximately 20,000 to 30,000 levels. This decoupling — rising SIP flows despite market volatility — demonstrates the maturation of investor behavior toward disciplined, long-term investing.

2.3 Folio Expansion and Investor Participation

The industry now services **26.12 crore folios**, underscoring the democratization of investing. The investor analytics reveal a total of **5,000 unique investors** within the project dataset, generating **33,000 transactions** with a total investment amount of **INR 4 billion** and an average transaction size of **INR 107,440**.

2.4 Category Dynamics

Table 2.1: Category-Level Inflow Analysis (INR Crore)

Category	2024	2025	Total
Liquid	3,46,328	1,04,947	4,51,275
Sectoral/Thematic	78,107	25,722	1,03,829
Flexi Cap	47,551	16,438	63,989
Large & Mid Cap	43,169	14,583	57,752
Mid Cap	41,116	14,196	55,312
Short Duration	41,859	13,671	55,530
Small Cap	34,204	12,392	46,596
Hybrid	29,711	9,157	38,868
Large Cap	19,449	6,184	25,633
Value/Contra	12,744	4,236	16,980
Gilt	7,753	2,642	10,395
ELSS	4,627	1,453	6,080
Total	**7,06,618**	**2,25,621**	**9,32,239**

The Liquid category's dominance (48.4% of total) reflects institutional treasury allocations, while Sectoral/Thematic's second-place positioning signals growing retail appetite for concentrated thematic bets. The Year-over-Year growth rate of **17.17%** confirms sustained industry momentum.

Chapter 3

Business Problem

Despite the rapid growth of the Indian mutual fund industry, several critical challenges persist that impair decision-making for retail investors, fund managers, and wealth advisors.

3.1 Opaque Fund Evaluation

With over 40 schemes across multiple fund houses and categories (Equity, Debt, Hybrid, Liquid, Gilt, ELSS, Sectoral/Thematic, and others), investors face information overload. Simple return comparisons ignore risk, expense ratios, and benchmark-relative performance. A fund delivering 23% three-year returns may appear superior to one delivering 15%, yet the former may carry a maximum drawdown of -52.57% (fund 119599) while the latter may have a drawdown of only -12.97% (fund 120843). Without risk-adjusted scoring, investors make suboptimal allocation decisions.

3.2 Absence of Risk-Adjusted Metrics

Retail investors rarely have access to Sharpe Ratios, Sortino Ratios, Alpha, Beta, or drawdown analysis. The project data reveals Sharpe Ratios ranging from -0.82 (fund 101208) to **1.31** (fund 120843), and Sortino Ratios from -1.68 (fund 101208) to **2.14** (fund 119551). These wide dispersions are invisible in standard return-only comparisons.

3.3 Investor Behavioral Blind Spots

Fund houses and advisors lack granular analytics on investor behavior — SIP continuity patterns, geographic concentration, age-cohort preferences, and transaction-type splits. The SIP continuity analysis identifies numerous investors with "At Risk" status, characterized by average gap days ranging from 36 to 93 days between SIP installments.

3.4 No Integrated Decision-Support System

Industry data, fund performance, investor analytics, and market trends exist in disconnected silos. A unified, interactive dashboard that enables filtering by fund house, category, date range, state, age group, and city tier — as developed in this project — addresses this fragmentation.

Chapter 4

Project Objectives

The project is organized around six clearly defined analytical objectives:

Objective 1: **Industry Analysis.** Analyze the macro-level Indian mutual fund landscape including AUM growth trends (2022–2025), SIP inflow trajectories, folio expansion, and fund house market share. Quantify the top 10 fund houses by AUM and identify category-level inflow patterns.

Objective 2: **Fund Performance Analysis.** Compute and analyze daily returns, annualized returns, CAGR, annualized standard deviation, and expense ratios for all 40 schemes. Develop a comprehensive performance scorecard integrating return, risk, and cost metrics.

Objective 3: **Risk Analytics.** Calculate Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value-at-Risk (95%), and Conditional VaR (95%) for each fund. Perform rolling 90-day Sharpe Ratio analysis to capture temporal risk-return dynamics.

Objective 4: **Investor Analytics.** Analyze investor demographics (age groups, states, city tiers), transaction behavior (SIP vs. Lumpsum vs. Redemption split), monthly transaction volumes, and SIP continuity patterns. Identify "At Risk" investors through gap-day analysis.

Objective 5: **Dashboard Development.** Build a four-page interactive Power BI dashboard with KPI cards, trend charts, comparative bar charts, scatter plots, bubble charts, donut charts, and data tables. Implement dynamic filters for fund house, category, plan, date range, state, age group, and city tier.

Objective 6: **Advanced Analytics & Recommendation Engine.** Develop a composite fund scoring model (0–100 scale) integrating return, Sharpe, Alpha, expense ratio, drawdown, and volatility. Perform sector concentration analysis using HHI. Build an investor cohort segmentation framework.

Chapter 5

Datasets & Data Engineering

5.1 Data Sources

Table 5.1: Project Datasets

Dataset	Description	Key Fields
Industry AUM	Quarterly AUM by fund house (2022–2025)	fund_house, aum, date
NAV History	Daily NAV for 40 schemes	amfi_code, date, nav
SIP Inflow Data	Monthly SIP inflows by category	category, sip_inflow_crore
Investor Transactions	Synthetic investor transaction records	investor_id, amount, type
Market Benchmark	Nifty 50 daily close values	date, close_value
Fund Metadata	Scheme details, category, plan	amfi_code, category, plan

5.2 Data Cleaning & Validation

A rigorous cleaning pipeline was implemented:

- **Missing Value Handling:** NAV histories were checked for gaps; missing trading days were forward-filled to maintain time-series continuity. Investor records with null state or age-group fields were flagged and imputed where possible.
- **Data Type Standardization:** Date fields were converted to datetime objects, numeric fields were cast to float64, and categorical fields were validated against controlled vocabularies.
- **Duplicate Removal:** Deduplication was performed on investor_id $\times$ transaction_date $\times$ amfi_code combinations to prevent inflated transaction counts.

- **Outlier Detection:** NAV values were screened for anomalous single-day jumps exceeding ±20%, which could indicate data errors or corporate actions. Such records were investigated and adjusted.

5.3 Feature Engineering

Several derived features were computed:

- **Daily Returns:** $(NAV_t - NAV_{t-1})/NAV_{t-1}$
- **Annualized Return:** Geometric mean of daily returns scaled to 252 trading days
- **Rolling Metrics:** 90-day rolling Sharpe Ratio, rolling standard deviation
- **Investor Gap Days:** Average number of days between consecutive SIP transactions per investor
- **SIP Continuity Status:** Classified as "At Risk" if average gap exceeds 30+ days
- **HHI:** Sum of squared sector allocation weights per fund
- **Fund Base 100 / Benchmark Base 100:** Rebased NAV and Nifty 50 series for normalized comparison

5.4 ETL Process

Data was ingested from CSV files into a PostgreSQL database via Python scripts. Transformation logic was implemented in Pandas, with computed metrics exported as dedicated CSV files (`sharpe_values.csv`, `sortino_values.csv`, `alpha_beta.csv`, `max_drawdown.csv`, `fund_scorecard.csv`, `sip_continuity.csv`). The Power BI dashboard connects to these processed outputs for visualization.

Chapter 6

Exploratory Data Analysis

6.1 Industry AUM Growth (2022–2025)

The AUM Growth Trend analysis reveals a compelling narrative of the Indian mutual fund industry's trajectory. Starting at approximately INR 82–85 lakh crore in early 2022, total industry AUM ascended gradually through 2022 and 2023, reaching approximately INR 95 lakh crore by mid-2023. The growth then accelerated sharply from mid-2023 through 2024, driven by a sustained equity bull market and record SIP inflows, peaking at approximately **INR 150 lakh crore** in late 2024. The subsequent decline to approximately INR 120 lakh crore by early 2025 corresponds to the global market correction triggered by geopolitical tensions, rising interest rate expectations, and profit-booking in mid and small-cap segments.

Business Interpretation: The non-linear growth pattern confirms that AUM is highly sensitive to equity market cycles, underscoring the need for category diversification (Debt, Liquid, Hybrid) to stabilize industry-level AUM during corrections.

6.2 Top 10 Fund Houses by AUM

Table 6.1: Top 10 Fund Houses by Assets Under Management

Rank	Fund House	Approx. AUM (INR Lakh Cr)
1	SBI Mutual Fund	∼13
2	ICICI Prudential MF	∼11
3	HDFC Mutual Fund	∼10
4	Nippon India MF	∼8
5	Kotak Mahindra MF	∼6
6	Aditya Birla Sun Life MF	∼5.5
7	UTI Mutual Fund	∼4.5
8	Axis Mutual Fund	∼4
9	Mirae Asset MF	∼3.5
10	DSP Mutual Fund	∼2

Strategic Implication: The top three fund houses command approximately 50% of total industry AUM, indicating high market concentration. New entrants face significant distribution and brand barriers. However, mid-tier players like Mirae Asset and DSP have carved niches in specific categories, suggesting that category specialization can be an effective growth strategy.

6.3 Age-Group Investment Patterns

Table 6.2: Average Investment by Age Group

Age Group	Average Investment (INR)
18–25	∼50,000
26–35	∼100,000
36–45	∼100,000
46–55	∼100,000
56+	∼100,000

The 18–25 cohort invests approximately half the amount of older cohorts, consistent with lower disposable incomes and shorter investment horizons. The relative parity among the 26–55+ cohorts suggests that once investors enter the workforce, investment capacity remains stable.

6.4 Transaction Type Distribution

Table 6.3: Transaction Type Split

Transaction Type	Share (%)	Amount (INR)
SIP	58.49	∼2 billion
Lumpsum	35.34	∼1 billion
Redemption	6.17	∼0 billion

SIP's dominance at 58.49% validates the industry's SIP-first growth strategy. The low redemption rate of 6.17% indicates strong investor stickiness within the analyzed period.

6.5 Risk vs. Return Analysis

The scatter plot analysis maps all 40 funds along risk (standard deviation, x-axis) and return (3-year return percentage, y-axis) dimensions, color-coded by category (Debt vs. Equity). Equity funds cluster in the 10–25% standard deviation range with returns of 12–28%, while Debt funds cluster near the origin with standard deviations below 10% and returns of 0–5%. Within equity, high-return outliers at approximately 23% return demonstrate that selective stock-picking or sectoral concentration can generate substantial alpha, albeit with proportionally higher volatility (25%+ standard deviation).

6.6 Geographic Distribution

Investment Amount by State reveals a notably uniform distribution across the top seven states: Punjab, Tamil Nadu, Madhya Pradesh, Rajasthan, Gujarat, West Bengal, and Telangana each contribute approximately INR 0.3 billion. This suggests broad geographic penetration of mutual fund investments across Tier 1 and Tier 2 cities.

6.7 Monthly Transaction Volume

The monthly transaction volume time-series spans from mid-2023 to early 2025, showing daily transaction counts fluctuating between 40 and 100, with a marked spike exceeding 80 transactions per day in early 2025. This surge likely correlates with tax-saving activity (ELSS investments before March 31) and new SIP registrations.

Chapter 7

Performance Analytics

For each of the 40 mutual fund schemes identified by AMFI code, the following metrics were computed using Python (Pandas, NumPy, SciPy) against the Nifty 50 as the benchmark.

7.1 Daily Returns

Daily logarithmic returns were computed as:

$$r_t = \ln\left(\frac{NAV_t}{NAV_{t-1}}\right) \tag{7.1}$$

The NAV trend analysis reveals that fund 100025 dominates in absolute NAV terms, peaking at approximately 1.05M in 2025 before declining sharply to 0.45M in 2026. Fund 100033 maintains a steady NAV trajectory at approximately 0.15–0.20M throughout, reflecting lower volatility.

7.2 Sharpe Ratio

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \tag{7.2}$$

Where R_p is annualized portfolio return, R_f is the risk-free rate, and σ_p is annualized standard deviation.

Table 7.1: Sharpe Ratio — Selected Funds

AMFI Code	Sharpe Ratio	Interpretation
119551	1.2083	Excellent
100033	1.0937	Strong
118632	1.0817	Strong
101206	1.0272	Strong
120504	1.0265	Strong
119094	0.9982	Near-Excellent
119552	0.9533	Good
119598	0.9453	Good
102885	0.8171	Good
120503	0.7980	Good
101208	−0.8156	Poor
100025	−0.5671	Poor

The average Sharpe Ratio across all funds is **0.54**. Funds with Sharpe Ratios above 1.0 consistently deliver excess returns per unit of risk. Fund 119551 leads at 1.21, while fund 101208 at −0.82 indicates returns fail to compensate for risk borne.

7.3 Sortino Ratio

$$\text{Sortino Ratio} = \frac{R_p - R_f}{\sigma_{downside}} \tag{7.3}$$

The Sortino Ratio penalizes only downside volatility, not upside deviation.

Table 7.2: Sortino Ratio — Selected Funds

AMFI Code	Sortino Ratio	Interpretation
119551	2.1403	Outstanding
118632	1.8501	Excellent
100033	1.8291	Excellent
120504	1.8053	Excellent
101206	1.7996	Excellent
119094	1.7038	Excellent
119598	1.6753	Excellent
119552	1.6098	Very Good
102885	1.4357	Good
120503	1.3379	Good
101208	−1.6810	Very Poor
100025	−0.9418	Poor

Fund 119551 achieves the highest Sortino of **2.14**, indicating exceptionally strong return generation relative to downside risk, effectively shielding investors from left-tail losses.

7.4 Alpha & Beta

$$\alpha = R_p - [R_f + \beta \times (R_m - R_f)] \tag{7.4}$$

$$\beta = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)} \tag{7.5}$$

Table 7.3: Alpha and Beta — Selected Funds

AMFI Code	Alpha	Beta	Interpretation
119598	0.2980	7.99E-06	High alpha, near-zero beta
100033	0.2585	2.31E-05	High alpha
119094	0.2546	7.07E-06	High alpha
119551	0.2267	7.39E-06	High alpha
118632	0.2203	−3.92E-06	High alpha, negative beta
118634	0.2061	−4.74E-05	High alpha, negative beta
120504	0.2017	1.83E-05	High alpha
100025	0.0427	2.70E-07	Low alpha
119599	0.0448	1.01E-05	Low alpha

The average Alpha across all funds is **0.16**. The near-zero Beta values across all funds indicate that daily return correlations between individual mutual fund NAVs and the Nifty 50 benchmark are extremely low at daily frequency — consistent with once-daily NAV repricing.

7.5 Maximum Drawdown

$$MDD = \frac{\text{Trough} - \text{Peak}}{\text{Peak}} \tag{7.6}$$

Table 7.4: Maximum Drawdown — Selected Funds

AMFI Code	Max DD (%)	Worst Date	Severity
119599	−52.57	28-Oct-2025	Extreme
119095	−51.68	11-May-2026	Extreme
101207	−35.45	11-May-2026	Severe
119598	−28.71	14-May-2025	Significant
102886	−28.00	27-Apr-2026	Significant
118634	−23.34	20-Feb-2026	Moderate
100033	−16.22	12-May-2022	Moderate
120843	−12.97	—	Low
101208	−0.16	12-Sep-2023	Negligible

Critical Observation: Fund 119599, despite having a return of 23.14%, suffered a catastrophic 52.57% drawdown — meaning an investor at peak would have lost more than half their capital before recovery. This extreme drawdown is directly responsible for its low composite score of 72.30.

7.6 Fund Scorecard

The composite fund scorecard integrates multiple metrics into a single 0–100 score, incorporating return_3yr_pct, sharpe_ratio, alpha, expense_ratio_pct, max_drawdown, and std_dev_ann_pct.

Table 7.5: Fund Scorecard — Top 15 Funds

Rank	AMFI	Score	3Y Ret%	Sharpe	Alpha	Exp%	Max DD%	Std%
1	120505	100.00	18.08	1.18	0.30	1.36	−18.19	19.29
2	119598	99.55	23.39	0.95	0.30	1.43	−28.71	25.14
3	149324	92.42	20.08	0.95	0.29	1.52	−31.17	24.84
4	100033	90.69	16.58	1.09	0.26	1.38	−16.22	18.94
5	120843	90.39	15.65	1.31	0.28	1.45	−12.97	15.89
6	119094	84.98	15.18	1.00	0.25	1.38	−20.96	19.41
7	148567	83.18	14.81	1.45	0.27	1.46	−11.27	14.19
8	149323	82.28	17.16	1.13	0.25	1.61	−17.25	17.75
9	148569	75.90	13.58	1.23	0.29	1.60	−16.40	17.67
10	118634	75.15	20.15	0.45	0.21	1.53	−23.34	25.24
11	118632	75.15	14.00	1.08	0.22	1.51	−17.41	14.15
12	120504	74.47	14.41	1.03	0.20	0.80	−12.59	14.36
13	119599	72.30	23.14	−0.06	0.04	0.72	−52.57	24.95
14	119551	69.74	12.36	1.21	0.23	1.54	−15.01	13.74
15	101207	69.44	22.38	0.16	0.10	1.53	−35.45	25.79

Key Insight: Fund 120505 achieves the perfect score of 100.00 not because it has the highest return (18.08% vs. 23.39% for fund 119598), but because it delivers the best *risk-adjusted balance* — a Sharpe of 1.18, a manageable drawdown of −18.19%, and a moderate standard deviation of 19.29%.

Chapter 8

Advanced Analytics

8.1 Historical VaR (95%) and CVaR (95%)

Value-at-Risk (VaR) at the 95% confidence level represents the maximum expected daily loss that will not be exceeded on 95% of trading days.

$$VaR_{95} = \text{Percentile}(\text{Daily Returns}, 5\%) \tag{8.1}$$

Conditional VaR (CVaR), also known as Expected Shortfall, measures the average loss in the worst 5% of scenarios:

$$CVaR_{95} = E[R \mid R \leq VaR_{95}] \tag{8.2}$$

The Maximum Drawdown data provides a complementary tail-risk perspective. Fund 119095 with a -51.68% max drawdown and fund 119599 with -52.57% represent the extreme left tail of the risk distribution. For conservative investors, funds like 120843 (max DD -12.97%) or 101208 (max DD -0.16%) represent significantly lower tail risk.

8.2 Rolling 90-Day Sharpe Ratio

The Rolling 90-Day Sharpe Ratio analysis tracks the time-varying risk-adjusted performance of the top five funds (120505, 119598, 149324, 100033, 120843) from mid-2022 through mid-2026.

Key Observations:

- **Fund 120505:** Starts with an exceptionally high Sharpe of approximately 5.8 in early 2022, then normalizes to fluctuate between 0 and 4. By early 2026, it stabilizes around 2–3, indicating consistent risk-adjusted performance.

- **Fund 120843:** Exhibits the most volatile rolling Sharpe, ranging from approximately -1 to 6.2. The spike to 6.2 in late 2024/early 2025 represents exceptional returns relative to risk.
- **Fund 100033:** Maintains a relatively stable rolling Sharpe between 1 and 5, with a notable peak to approximately 5 in early 2026.
- **Fund 149324:** The most volatile performer, with the rolling Sharpe dropping to approximately -2.5 in late 2023, indicating severe underperformance.
- **Fund 119598:** Displays moderate volatility with the rolling Sharpe generally between -1 and 4, showing cyclical performance patterns.

Business Interpretation: Rolling Sharpe analysis reveals that static, point-in-time metrics can mask significant temporal variation in fund quality. This analysis is critical for tactical allocation decisions and for identifying funds with consistent versus cyclical alpha generation.

8.3 SIP Continuity Analysis

Investors are classified based on the average gap between consecutive SIP installments:

Table 8.1: SIP Continuity — At Risk Investors (Sample)

Investor ID	Avg Gap Days	Status
INV000028	93.6	At Risk
INV000004	85.4	At Risk
INV000072	83.0	At Risk
INV000014	75.3	At Risk
INV000056	74.8	At Risk
INV000008	70.4	At Risk
INV000045	70.6	At Risk
INV000010	64.8	At Risk
INV000029	60.7	At Risk
INV000012	57.0	At Risk
INV000052	36.2	At Risk
INV000011	40.2	At Risk

Investors with average gap days exceeding 30 are flagged as “At Risk” of SIP discontinuation. INV000028 with a 93.6-day average gap is the most likely to churn — nearly quarterly gaps between installments suggest disengagement. Fund houses should implement targeted re-engagement campaigns for these investors.

8.4 Sector Concentration Analysis (HHI)

The Herfindahl-Hirschman Index (HHI) measures portfolio concentration:

$$HHI = \sum_{i=1}^{n} s_i^2 \tag{8.3}$$

Where s_i is the portfolio weight of sector i.

Table 8.2: Top Concentrated Funds by HHI

Rank	AMFI Code	HHI	Concentration Level
1	119092	∼0.205	Highly Concentrated
2	101207	∼0.200	Highly Concentrated
3	119599	∼0.175	Moderate-High
4	102885	∼0.175	Moderate-High
5	118632	∼0.170	Moderate
6	148568	∼0.170	Moderate
7	120505	∼0.160	Moderate
8	120506	∼0.155	Moderate
9	125498	∼0.155	Moderate
10	120841	∼0.150	Moderate

Cross-Reference: Fund 119092 (highest HHI at ∼0.205) has a Sharpe of only 0.03, suggesting extreme concentration does not translate into superior risk-adjusted returns. Fund 120505 (rank 1 by composite score) has a moderate HHI of ∼0.160, suggesting optimal performance emerges from moderate sector concentration.

8.5 Fund Recommendation Engine

The composite fund scorecard serves as the recommendation engine:

Table 8.3: Fund Recommendation Tiers

Tier	Score	Funds	Action
Tier 1 — Strong Buy	90–100	120505, 119598, 149324, 100033, 120843	Core holdings
Tier 2 — Buy	75–89	119094, 148567, 149323, 148569, 118634, 118632, 120504	Tactical allocation
Tier 3 — Hold	60–74	119599, 119551, 101207, 148568, 120506	Monitor closely
Tier 4 — Review	<60	Remaining funds	Replace

Chapter 9

Power BI Dashboard

The interactive Power BI dashboard comprises four pages, each serving a distinct analytical purpose.

9.1 Page 1 — Industry Overview

Table 9.1: Industry Overview KPIs

KPI	Value
Total AUM	INR 62.74 lakh crore
SIP Inflows	INR 31K crore (monthly)
Folios	26.12 crore
Total Schemes	40

Visualizations: AUM Growth Trend line chart (2022–2025) depicting the rise from INR 82 lakh crore to INR 150 lakh crore followed by correction. Top 10 Fund Houses horizontal bar chart ranking SBI, ICICI Prudential, HDFC, Nippon India, Kotak, ABSL, UTI, Axis, Mirae Asset, and DSP by AUM. Dynamic filters include Fund House selector and Date range slider (31-03-2022 to 31-12-2025).

9.2 Page 2 — Fund Performance

Table 9.2: Fund Performance KPIs

KPI	Value
Avg 3Y Return	14.09%
Avg Sharpe	0.54
Avg Alpha	0.16
Top Score	100.00

Visualizations: Risk vs. Return scatter plot mapping all 40 funds by standard deviation and return, color-coded by category (Debt/Equity). Fund Scorecard data table displaying fund_rank, amfi_code, score_0_100, return_3yr_pct, sharpe_ratio, alpha, and expense_ratio_pct. NAV vs. Benchmark line chart comparing Fund Base 100 and Benchmark Base 100. Filters include Fund House, Category, and Plan selectors.

9.3 Page 3 — Investor Analytics

Table 9.3: Investor Analytics KPIs

KPI	Value
Total Invest Amount	INR 4 billion
Avg Transaction Amount	INR 107.44K
Total Investors	5,000
Total Transactions	33,000

Visualizations: Investment Amount by State horizontal bar chart, Average Investment by Age Group bar chart, Monthly Transaction Volume time-series, SIP vs. Lumpsum vs. Redemption donut chart (58.49%/35.34%/6.17%). Filters include State, Age Group, and City Tier selectors.

9.4 Page 4 — SIP & Market Trends

Table 9.4: SIP & Market Trends KPIs

KPI	Value
Latest YoY Growth	17.17%
Latest Active SIP Account	9.35 crore
Latest SIP AUM	INR 15.90 lakh crore

Visualizations: Top 5 Categories by Net Inflow horizontal bar chart (Liquid dominates at $\sim$0.5M crore). SIP Inflows vs. Nifty 50 dual-axis chart with SIP inflows growing from INR 0.13M in 2022 to INR 0.35M in 2025. Comprehensive category inflow table spanning 12 categories. Filters include Category selector and Date range slider.

Chapter 10

Key Findings

Based on exhaustive analysis of all project outputs, the following evidence-based findings are established:

Finding 1: Fund 120505 achieves the highest composite score of **100.00**, driven by a balanced combination of 18.08% three-year return, 1.18 Sharpe Ratio, 0.30 Alpha, and a manageable −18.19% maximum drawdown.

Finding 2: Extreme returns do not guarantee top rankings. Fund 119599 delivers 23.14% three-year return but ranks only 13th with a score of 72.30 due to a catastrophic −52.57% maximum drawdown and a negative Sharpe Ratio of −0.06.

Finding 3: SIP transactions constitute **58.49%** of all investment flows, with Lumpsum at 35.34% and Redemptions at only 6.17%, demonstrating that systematic investing dominates investor behavior.

Finding 4: The Liquid category commands **48.4%** of total inflows (INR 4,51,275 crore out of INR 9,32,239 crore), reflecting institutional treasury allocations.

Finding 5: The top three fund houses (SBI, ICICI Prudential, HDFC) collectively control approximately **50%** of industry AUM, indicating significant market concentration.

Finding 6: Rolling 90-day Sharpe Ratio analysis reveals that even the top-ranked fund (120505) experiences periods where its rolling Sharpe drops below 1.0, underscoring that static metrics can mask temporal performance variation.

Finding 7: SIP continuity analysis identifies multiple investors with average gap days exceeding 60 (e.g., INV000028 at 93.6 days), flagging them as "At Risk" for SIP discontinuation.

Finding 8: Sector concentration analysis via HHI reveals that funds 119092 (HHI ∼0.205) and 101207 (HHI ∼0.200) are highly concentrated, and this correlates with poor risk-adjusted metrics.

Finding 9: The 18–25 age cohort invests approximately **50% less** (∼INR 50K average) than older cohorts (∼INR 100K average), representing an underserved demographic.

Finding 10: Year-over-Year industry growth of **17.17%**, combined with 9.35 crore active SIP accounts and INR 15.90 lakh crore SIP AUM, confirms the structural shift toward disciplined investing.

Finding 11: Fund 120843 achieves the highest Sharpe Ratio of **1.31** and the lowest maximum drawdown among equity funds at −12.97%, positioning it as the optimal risk-minimizing equity allocation.

Finding 12: Beta values across all funds are near-zero at daily frequency (ranging from −4.74E-05 to 2.31E-05), suggesting the need for category-specific benchmarks rather than a single broad-market index.

Chapter 11

Business Recommendations

11.1 For Retail Investors

1. **Prioritize risk-adjusted metrics over raw returns.** The fund scorecard demonstrates that funds ranking highly on returns (e.g., 119599 at 23.14%) may carry unacceptable drawdown risk (-52.57%). Use Sharpe Ratio (target > 1.0) and Maximum Drawdown (target $< -20\%$) as primary screening criteria.

2. **Maintain SIP discipline.** The SIP continuity analysis shows that gap days of 60+ indicate likely discontinuation. Set up auto-debit mandates and resist pausing SIPs during corrections.

3. **Diversify across categories.** A blended allocation across 2–3 categories with uncorrelated return streams reduces portfolio-level drawdown.

11.2 For Fund Managers

4. **Monitor rolling Sharpe for consistency.** Implement internal rolling-risk dashboards to detect deterioration early and adjust portfolio positioning proactively.

5. **Manage sector concentration.** Funds with HHI > 0.18 demonstrate poor risk-adjusted outcomes. Rebalancing toward moderate concentration (HHI 0.12–0.16) optimizes the risk-return trade-off.

11.3 For Asset Management Companies

6. **Implement churn prediction models.** The SIP continuity data provides a natural training set for machine learning classifiers that predict investor discontinuation.

7. **Target the 18–25 demographic.** Micro-SIP products (INR 100–500/month), gamified investment apps, and campus ambassador programs can capture early adoption.

11.4 For Wealth Advisors

8. **Use the fund scorecard as a client communication tool.** The 0–100 scoring framework simplifies complex multi-metric comparisons into an intuitive ranking.

9. **Leverage geographic analytics for market expansion.** The uniform distribution across multiple states suggests Tier 2 and Tier 3 cities represent untapped growth opportunities.

10. **Align recommendations with investor risk capacity.** The drawdown analysis provides concrete worst-case scenarios for expectation management.

Chapter 12

Project Limitations

1. **Historical Data Dependency.** All performance metrics are backward-looking. Past performance does not predict future returns, and regime changes can invalidate historical relationships.

2. **Synthetic Investor Data.** The investor transaction dataset is synthetically generated, meaning geographic distributions and behavioral patterns may not accurately reflect actual investor behavior.

3. **Single Benchmark Limitation.** All Alpha and Beta calculations use the Nifty 50 as the sole benchmark. Debt, Liquid, and Hybrid funds should ideally use category-specific benchmarks.

4. **Daily Beta Near-Zero Issue.** The near-zero Beta values at daily frequency limit interpretability of CAPM-based Alpha. Monthly intervals would yield more meaningful estimates.

5. **Static Expense Ratios.** The analysis uses point-in-time expense ratios, which may vary over time as AUM changes.

6. **Survivorship Bias.** The dataset includes only currently active schemes, excluding funds that may have been merged or wound up, creating an upward bias in average performance.

7. **Limited Temporal Coverage.** While data spans 2022–2026, a longer history (10+ years) would enable more robust long-term trend analysis.

Chapter 13

Future Enhancements

1. **Machine Learning NAV Forecasting.** Implement LSTM (Long Short-Term Memory) and ARIMA models to forecast NAV trajectories for 30/60/90-day horizons.
2. **Real-Time NAV Tracking.** Integrate AMFI's daily NAV API to enable live dashboard updates and real-time portfolio valuation.
3. **Churn Prediction Model.** Train a supervised classification model (Random Forest, XGBoost) on SIP continuity data to predict investor churn probability with SHAP explainability.
4. **Portfolio Optimization Engine.** Implement Markowitz Mean-Variance Optimization and Black-Litterman models to generate optimal multi-fund portfolios along the efficient frontier.
5. **AI-Powered Recommendation System.** Develop collaborative filtering or content-based recommendation engine suggesting funds based on investor demographics and risk profile.
6. **NLP-Based Market Sentiment Analysis.** Ingest financial news and RBI policy statements through NLP pipelines to generate sentiment scores.
7. **Automated Report Generation.** Build a Python-based reporting pipeline that auto-generates PDF fund fact sheets monthly.
8. **Mobile Dashboard.** Deploy via Power BI Mobile or develop a custom Flask/Streamlit web application for cross-platform accessibility.

Chapter 14

Conclusion

The Bluestock Mutual Fund Analytics & Investor Insights Platform represents a comprehensive, end-to-end data analytics initiative that transforms fragmented financial data into structured, actionable intelligence for the Indian mutual fund ecosystem.

14.1 Project Achievements

- Successfully ingested, cleaned, and engineered features from multiple datasets spanning industry-level AUM, scheme-level NAV histories, investor transactions, and market benchmarks.
- Computed a complete suite of risk-adjusted performance metrics — Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown — for all 40 schemes, revealing that fund 120505 ranks first with a perfect composite score of **100.00**.
- Developed advanced analytics including rolling 90-day Sharpe Ratio analysis, SIP continuity modeling (identifying "At Risk" investors with gap days up to 93.6), sector concentration analysis via HHI, and VaR/CVaR tail-risk quantification.
- Built a four-page interactive Power BI dashboard covering Industry Overview (AUM INR 62.74 lakh crore, 26.12 crore folios), Fund Performance (Avg 3Y Return 14.09%, Avg Sharpe 0.54), Investor Analytics (INR 4 billion total investments, 58.49% SIP share), and SIP & Market Trends (17.17% YoY growth).
- Generated 12 evidence-based findings and 10 actionable business recommendations grounded in project-specific analytical outputs.

14.2 Business Value

The platform provides a replicable analytical framework that can be deployed by fund houses, wealth advisory firms, and fintech platforms to evaluate fund quality, segment investors, predict churn, and optimize portfolios. The composite fund scorecard alone — by integrating return, risk, alpha, cost, and drawdown into a single 0–100 metric — eliminates the complexity that prevents retail investors from making informed allocation decisions.

14.3 Learning Outcomes

This project demonstrates proficiency in data engineering (ETL, cleaning, feature engineering), quantitative finance (risk-adjusted metrics, CAPM, VaR), statistical analysis (rolling windows, correlation, HHI), business intelligence (Power BI dashboard design), and professional technical documentation — collectively representing the full spectrum of competencies expected of a data analytics professional operating at the intersection of finance and technology.

Appendix A

Formulas Reference

A.1 Compound Annual Growth Rate (CAGR)

$$CAGR = \left(\frac{V_{final}}{V_{initial}}\right)^{\frac{1}{n}} - 1 \tag{A.1}$$

Where V_{final} is the ending NAV, $V_{initial}$ is the starting NAV, and n is the number of years.

A.2 Sharpe Ratio

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p} \tag{A.2}$$

Where R_p is annualized portfolio return, R_f is the risk-free rate, and σ_p is annualized standard deviation.

A.3 Sortino Ratio

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_{downside}} \tag{A.3}$$

Where $\sigma_{downside}$ is the standard deviation of negative returns only.

A.4 Alpha (Jensen's Alpha)

$$\alpha = R_p - [R_f + \beta \times (R_m - R_f)] \tag{A.4}$$

Where R_m is the benchmark return and β is the fund's systematic risk coefficient.

A.5 Beta

$$\beta = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)} \tag{A.5}$$

A.6 Maximum Drawdown

$$MDD = \frac{\text{Trough} - \text{Peak}}{\text{Peak}} \tag{A.6}$$

Measures the largest peak-to-trough decline in NAV.

A.7 Value-at-Risk (Historical, 95%)

$$VaR_{95} = \text{Percentile}(R, 5\%) \tag{A.7}$$

The 5th percentile of the historical daily return distribution.

A.8 Conditional VaR (Expected Shortfall, 95%)

$$CVaR_{95} = E[R \mid R \leq VaR_{95}] \tag{A.8}$$

The mean of all returns falling below the VaR threshold.

A.9 Herfindahl-Hirschman Index (HHI)

$$HHI = \sum_{i=1}^{n} s_i^2 \tag{A.9}$$

Where s_i is the portfolio weight of sector i. HHI = 1 indicates complete concentration; HHI $\rightarrow$ 0 indicates perfect diversification.

End of Report

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